# Lab Eleven – Point Estimation and Resampling

**Tips**

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!
- Make sure to plot all computed probabilities. This is good `ggplot` practice AND will help make sure you don't make a mistake when computing the probability.

# 1 Introduction

When performing point estimation, there isn't just one *right* answer. For example, the Method of Moments (MOM) and Maximum Likelihood Estimates (MLEs) aren't always the same.

One question that comes up, is why we calculate the sample variance as

$$S^2_{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

when it is supposed to be the "average squared distance from the mean, i.e.,

$$S^2_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

# 2 Comparing Estimators

## 2.1 Initial Simulation

In this lab, we will explore why we use $S^2_{n-1}$ and not $S^2_n$. To do this, we will generate data from four known distributions:

- Poisson$(\lambda = 2)$
- Binomial$\left(n = 50, p = \frac{5-\sqrt{21}}{10}\right)$
- Normal$(\mu = 0, \sigma = \sqrt{2})$
- Exponential$\left(\lambda = \frac{1}{\sqrt{2}}\right)$.

**Remark**: Note that I have selected the parameters so that the true variance for all four models is 2.

For each distribution, write a loop that completes the following 1000 times. Put `set.seed(7272)` at the top of your code chunk.

- Generate a sample of size $n$ from the distribution.
- Calculate and save $S^2_{n-1}$ and $S^2_n$.

Start with $n = 20$. Is there evidence that $S^2_{n-1}$ is a better estimator than $S^2_n$?

Create a plot or combination of plots (using patchwork) to show the results across all four distributions. Make sure to create a corresponding table that numerically summarizes the results of the simulation. You should include

$$\text{Bias} = \text{Average of Estimates} - \text{True Value}$$
$$\text{Precision} = \frac{1}{\text{Variance of Estimates}}$$
$$\text{Mean Square Error} = \text{Bias}^2 + \text{Variance of Estimates}.$$

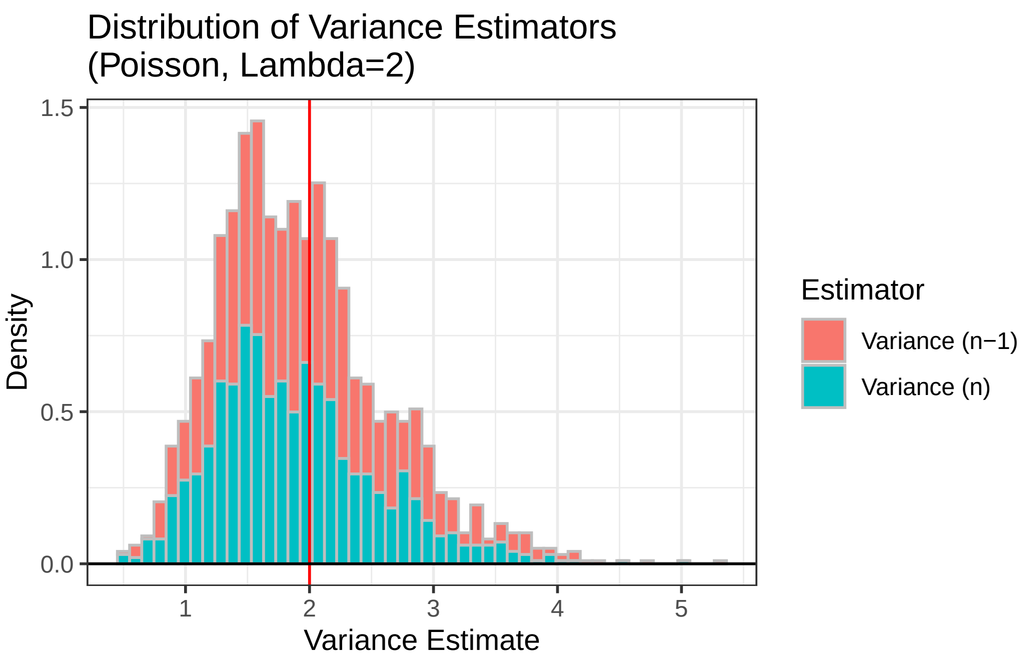
**Solution:**
*Poisson:*

```r
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
var.true = 2

for(i in 1:1000){
  x = rpois(n = n, lambda = 2)
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
poisson.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

poisson.comparison |>
  knitr::kable(digits = 2)
```

| Estimator | Bias | Precision | MSE |
|---|---|---|---|
| Variance (n-1) | -0.02 | 2.10 | 0.48 |
| Variance (n) | -0.12 | 2.33 | 0.44 |

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
```

```
)

ggdat |> ggplot() +
  geom_histogram(aes(x = value,
                     y = after_stat(density),
                     fill = estimator),
                     bins = 50,
                     color = "gray") +
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       fill = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Poisson, Lambda=2)")
```



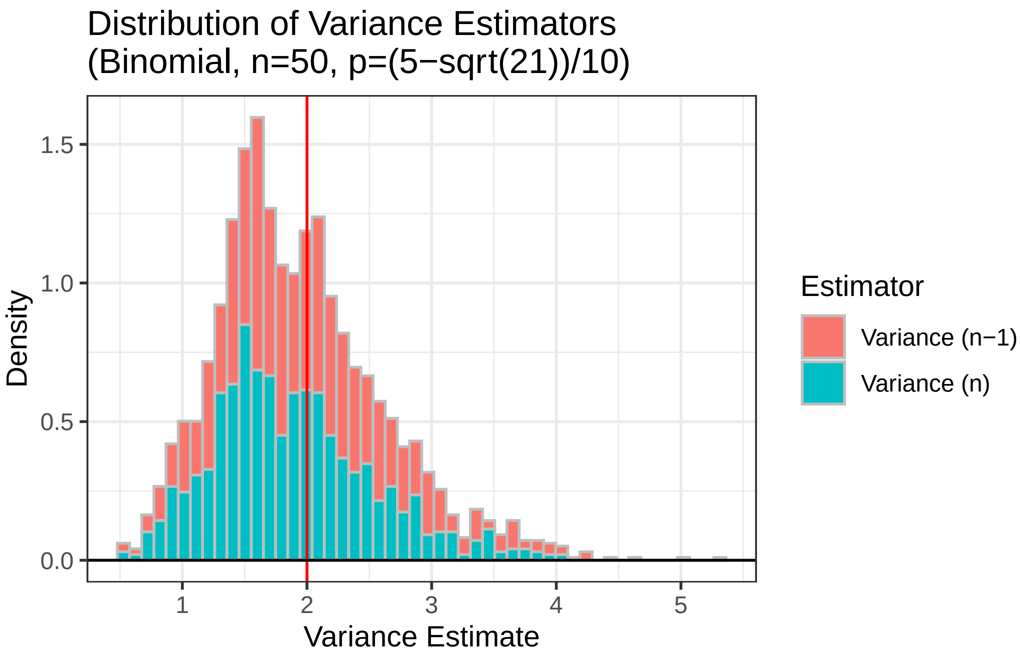

*Binomial:*

```
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
var.true = 2

for(i in 1:1000){
  x = rbinom(n,50,(5-sqrt(21))/10)
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
```

```r
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
binom.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

binom.comparison |>
  knitr::kable(digits = 2)
```

| Estimator | Bias | Precision | MSE |
|---|---|---|---|
| Variance (n-1) | -0.01 | 2.13 | 0.47 |
| Variance (n) | -0.11 | 2.36 | 0.44 |

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
)

ggdat |> ggplot() +
  geom_histogram(aes(x = value,
                     y = after_stat(density),
                     fill = estimator),
                     bins = 50,
                     color = "gray") +
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       fill = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Binomial, n=50, p=(5-sqrt(21))/10)")
```



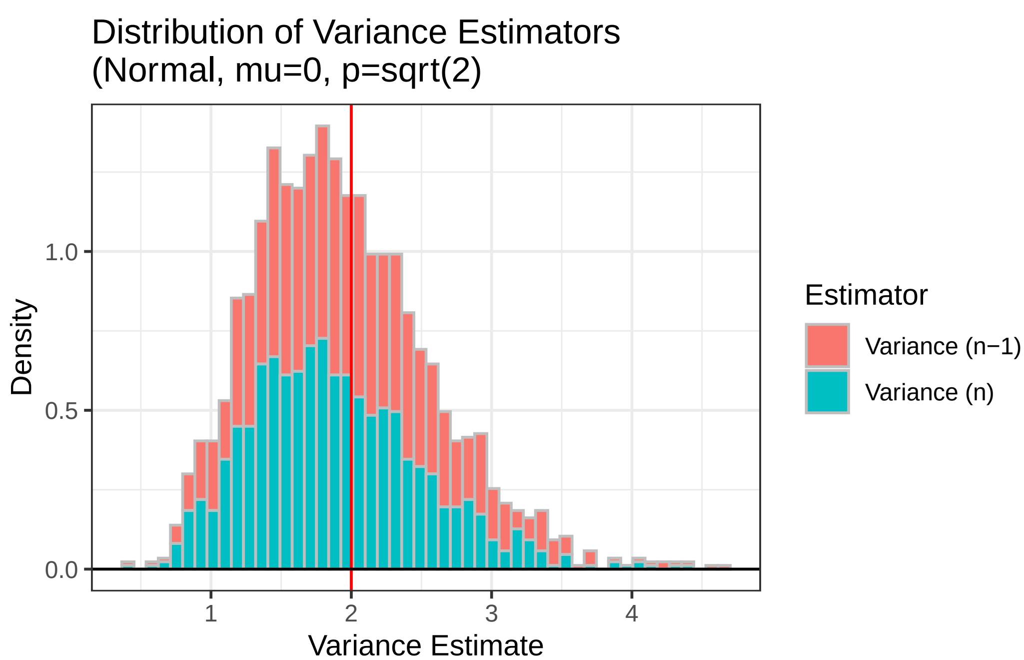

*Normal:*

```
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
var.true = 2

for(i in 1:1000){
  x = rnorm(n,0,sqrt(2))
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
norm.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

```

```r
norm.comparison |>
  knitr::kable(digits = 2)
```

| Estimator | Bias | Precision | MSE |
|---|---|---|---|
| Variance (n-1) | -0.01 | 2.41 | 0.41 |
| Variance (n) | -0.11 | 2.67 | 0.39 |

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
)

ggdat |> ggplot() +
  geom_histogram(aes(x = value,
                     y = after_stat(density),
                     fill = estimator),
                     bins = 50,
                     color = "gray") +
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       fill = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Normal, mu=0, p=sqrt(2)")
```



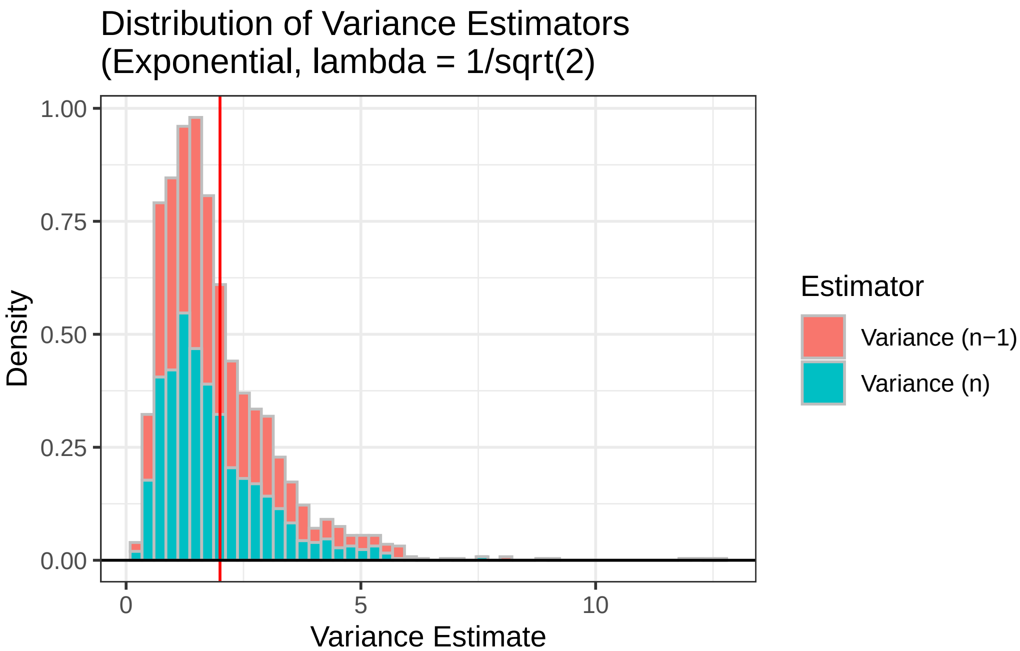

*Exponential:*

```r
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
```

```r
var.true = 2

for(i in 1:1000){
  x = rexp(n,rate=1/sqrt(2))
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
exp.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

exp.comparison |>
  knitr::kable(digits = 2)
```

| Estimator | Bias | Precision | MSE |
|---|---|---|---|
| Variance (n-1) | -0.03 | 0.61 | 1.64 |
| Variance (n) | -0.12 | 0.68 | 1.49 |

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
)

ggdat |> ggplot() +
  geom_histogram(aes(x = value,
                     y = after_stat(density),
                     fill = estimator),
                     bins = 50,
                     color = "gray") +
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       fill = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Exponential, lambda = 1/sqrt(2)")
```




### 2.2 Simulation over various sample sizes

Consider the effect of the sample size have on these results. Provide a graphical and numerical summary of what happens as $n$ increases – say $n = 20$, $n = 50$, $n = 100$, and $n = 500$. Does the result in your initial simulation hold across samples? Or, do things change as the sample size changes? Ensure to set your randomization seed.

### 2.3 Simulation

You have written code to draw a random sample of various sizes from each distribution, and to calculate and store the calculated estimates. Rework your code to produce a density plot of the $s^2_{n-1}$ and $s^2_n$. Do you have a guess about their distribution? Ensure to set your randomization seed.

### 2.4 Bootstrap

Now, take *one* sample from each distribution. Instead of drawing a random sample at each iteration, sample *with replacement* from the one sample. Produce a density plot of the $s^2_{n-1}$ and $s^2_n$. Do we get roughly the same information as we do in the full simulation? Try this over several seeds – what do you notice? Ensure to set your randomization seed.

## 3 Executive Summary

Summarize the work you've done here to provide a brief (200-300 word) summary of the simulation and the results using your work in Lab 11. Your summary should be professional, clear, and all claims should be corroborated with statistical support.